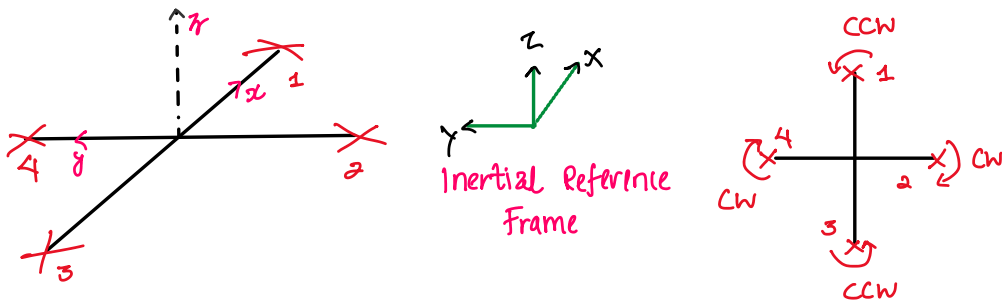




ZYX Euler Angle Rotation (Roll-Pitch-Yaw)

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \quad \text{Rotation matrix}$$

$$R_{zyx} = R_z(\psi) \cdot R_y(\theta) \cdot R_x(\phi)$$

$$R_{zyx} = \begin{bmatrix} \cos\psi \cos\theta & \cos\psi \sin\theta \sin\phi - \sin\psi \cos\phi & \cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi \\ \sin\psi \cos\theta & \sin\psi \sin\theta \sin\phi + \cos\psi \cos\phi & \sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi \\ -\sin\theta & \cos\theta \sin\phi & \cos\theta \cos\phi \end{bmatrix}$$

$$T_{BFR} = \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}, \quad T_{IFR} = R_{B \rightarrow I} \cdot T_{BFR} = R_{zyx} \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}$$

$$= T \begin{bmatrix} \cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi \\ \sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi \\ \cos\theta \cos\phi \end{bmatrix}$$

EOM

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = T \begin{bmatrix} \cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi \\ \sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi \\ \cos\theta \cos\phi \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} - A_{drag} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}$$

Translational Dynamics

$$\left. \begin{aligned} \ddot{x} &= \frac{I}{m} (c\phi \cdot s\theta \cdot c\psi + s\phi \cdot s\psi) - \frac{A_x}{m} \cdot \dot{x} \\ \ddot{y} &= \frac{I}{m} (c\phi \cdot s\theta \cdot s\psi - s\phi \cdot c\psi) - \frac{A_y}{m} \cdot \dot{y} \\ \ddot{z} &= \frac{I}{m} c\phi c\theta - \frac{A_z}{m} \cdot \dot{z} \cdot g \end{aligned} \right\} \rightarrow \frac{1}{s} \rightarrow \frac{1}{s} \rightarrow \xi \quad (A)$$

We get acceleration by solving three equations and integrate to get velocities and feed back to calculate air drag.

Rotational Dynamics

$$\begin{aligned} \textcircled{a} \Sigma M &= I_\phi \hat{i}_3 + I_\theta \hat{j}_3 + I_\psi \hat{k}_3 \quad (\text{Body axis}) \\ &= \underbrace{IK(\omega_4^2 - \omega_2^2)}_{\text{Roll torque}} \hat{i}_3 + \underbrace{IK(\omega_3^2 - \omega_1^2)}_{\text{Pitch torque}} \hat{j}_3 + \underbrace{b(\omega_4^2 + \omega_2^2 - \omega_1^2 - \omega_3^2)}_{\text{Yaw torque}} \hat{k}_3 \end{aligned}$$

$$\textcircled{b} \Sigma M = \frac{B}{dt} \vec{H} + \vec{\omega} \times \vec{H}$$

$$\textcircled{c} \vec{H} = I \cdot \vec{\omega} = I_{xx} \cdot p \hat{i}_3 + I_{yy} \cdot q \hat{j}_3 + I_{zz} \cdot r \hat{k}_3$$

What are p, q, & r? $\rightarrow$ Body rates about x, y, z of body or R_3 or BFR

$$\textcircled{1} \text{ So, when QC rotate about } z \rightarrow \vec{\omega} \xrightarrow{R \rightarrow R_1} \hat{K} = \hat{K}_1$$

$$\begin{bmatrix} \hat{i}_1 \\ \hat{j}_1 \end{bmatrix} = \begin{bmatrix} c\psi & -s\psi \\ s\psi & c\psi \end{bmatrix} \begin{bmatrix} \hat{i}_3 \\ \hat{j}_3 \end{bmatrix} \quad \psi \Rightarrow \text{Yaw}$$

$$\textcircled{2} \text{ when QC rotates about } y \rightarrow \vec{\omega} \xrightarrow{R_1 \rightarrow R_2} \hat{j}_1 = \hat{j}_2$$

$$\begin{bmatrix} \hat{k}_1 \\ \hat{i}_1 \end{bmatrix} = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix} \begin{bmatrix} \hat{k}_2 \\ \hat{i}_2 \end{bmatrix} \quad \theta \Rightarrow \text{Pitch}$$

$$\textcircled{3} \text{ when QC rotates about } x \rightarrow \vec{\omega} \xrightarrow{R_2 \rightarrow R_3} \hat{i}_2 = \hat{i}_3$$

$$\begin{bmatrix} \hat{j}_2 \\ \hat{k}_2 \end{bmatrix} = \begin{bmatrix} c\phi & -s\phi \\ s\phi & c\phi \end{bmatrix} \begin{bmatrix} \hat{j}_3 \\ \hat{k}_3 \end{bmatrix} \quad \phi \Rightarrow \text{Roll}$$

we know $\vec{R} \rightarrow \vec{R}_3 = \vec{R} \rightarrow \vec{R}_1 + \vec{R}_1 \rightarrow \vec{R}_2 + \vec{R}_2 \rightarrow \vec{R}_3$

$$= \hat{K}_1 \dot{\psi} + \hat{J}_2 \cdot \dot{\theta} + \hat{L}_3 \cdot \dot{\phi}$$

$$\downarrow \quad \quad \downarrow$$

$$\hat{K}_3 \quad \quad \hat{J}_2$$

Get $K_1 \rightarrow K_2$ from (a) & $K_2 \rightarrow K_3$ from (b)

$$(h) \vec{R} \rightarrow \vec{R}_3 = \underbrace{(\dot{\phi} - \dot{\psi} \sin \theta)}_p \hat{i}_3 + \underbrace{(\dot{\theta} \cos \phi + \dot{\psi} (\theta \cdot \sin \phi))}_q \hat{j}_3 + \underbrace{(\dot{\psi} (\theta \cos \phi - \dot{\theta} \sin \phi))}_r \hat{k}_3$$

By replacing (a) on (c) & (b) and equating, we get

$$LK(\omega_4^2 - \omega_2^2) = I_{x3} (\ddot{\phi} - \dot{\psi} \dot{\theta} \sin \theta - \dot{\psi} \dot{\theta} \cos \theta) + q r (I_{x3} - I_{y3})$$

$$LK(\omega_3^2 - \omega_4^2) = I_{y3} (\ddot{\theta} \cos \phi - \dot{\theta} \dot{\phi} \sin \phi + \ddot{\psi} (\theta \sin \phi - \dot{\psi} \dot{\theta} \sin \theta \sin \phi + \dot{\psi} \dot{\phi} \cos \phi) + p r (I_{x3} - I_{z3})$$

$$b(\omega_4^2 + \omega_2^2 - \omega_3^2 - \omega_1^2) = I_{z3} (-\ddot{\theta} \sin \phi - \dot{\theta} \dot{\phi} \cos \phi + \ddot{\psi} \cos \phi \cos \phi - \dot{\psi} \dot{\theta} \sin \theta \cos \phi - \dot{\psi} \dot{\theta} (\theta \cdot \sin \phi) + p q (I_{y3} - I_{x3}))$$

These eqn can be compressed down to three eqn.

$$J \frac{d^2 \eta}{dt^2} + C \frac{d\eta}{dt} = B \quad \text{--- (X)}$$

where, $\eta = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$, $J = 3 \times 3$ matrix = $J(\eta) \rightarrow$ Generalized mass eqn.
 $C = 3 \times 3$ matrix = $C(\eta, \dot{\eta}) \rightarrow$ Coriolis centripetal matrix

$$B = 3 \times 1 \text{ matrix} = [C_\phi \ C_\theta \ C_\psi]^T$$

We solve (X) to get $\ddot{\eta} \rightarrow \frac{1}{s} \rightarrow \frac{1}{s} \rightarrow \eta$

Inverse Mixer

we have

$$\left. \begin{aligned} \vec{T} &= k(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \vec{b}_3 \\ \vec{\tau}_\phi &= k(\omega_1^2 - \omega_2^2) \hat{b}_1 \\ \vec{\tau}_\theta &= k(\omega_3^2 - \omega_4^2) \hat{b}_2 \\ \vec{\tau}_\psi &= b(\omega_1^2 - \omega_2^2 - \omega_3^2 - \omega_4^2) \hat{b}_3 \end{aligned} \right\} \quad (c)$$

(c) Translates to

$$\left. \begin{aligned} \omega_1^2 &= \frac{T}{4k} - \frac{\tau_\theta}{2kl} - \frac{\tau_\psi}{4b} \\ \omega_2^2 &= \frac{T}{4k} - \frac{\tau_\phi}{2kl} + \frac{\tau_\psi}{4b} \\ \omega_3^2 &= \frac{T}{4k} + \frac{\tau_\theta}{2kl} - \frac{\tau_\psi}{4b} \\ \omega_4^2 &= \frac{T}{4k} + \frac{\tau_\phi}{2kl} + \frac{\tau_\psi}{4b} \end{aligned} \right\} \quad \text{This is what we need to calculate to get angular speeds of motor}$$

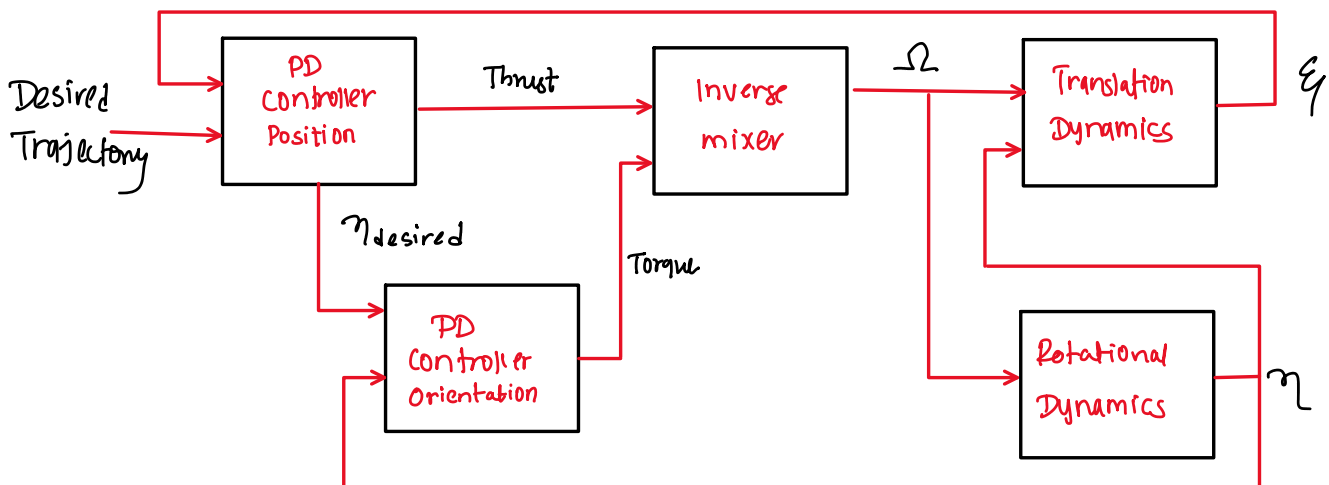


Fig: General overview of PD controller for path following GC (Trajectory follow)

1.Position and Orientation		
Variables	Body Frame [R ₃]	Inertial Frame [R ₁]
Position	-	$\zeta = [x \ y \ z]^T$
Roll Angle	Same in both frames	ϕ
Pitch Angle	Same in both frames	θ
Yaw Angle	Same in both frames	ψ
Euler and Vector	Same in both frames	$\eta = [\phi \ \theta \ \psi]^T$
2.Linear Velocity		
Velocity along x-axis	u	$\dot{x}$
Velocity along y-axis	v	$\dot{y}$
Velocity along z-axis	w	$\dot{z}$
Velocity vector	$V_b = [u \ v \ w]^T$	$\dot{\zeta} = [\dot{x} \ \dot{y} \ \dot{z}]$
Frame relation	$V_b = R \cdot \zeta$	$\dot{\zeta} = R^T \cdot V_b$
3.Linear Acceleration		
Acceleration along x-axis	$\dot{u} + wq - vr$	$\ddot{x}$
Acceleration along y-axis	$\dot{v} - wp + ur$	$\ddot{y}$
Acceleration along z-axis	$\dot{w} + vp - uq$	$\ddot{z}$
Note	Coupling $\omega \times V_b$	---
4.Angular Velocity		
Roll rate	p	$\dot{\phi}$
Pitch rate	q	$\dot{\theta}$
Yaw rate	r	$\dot{\psi}$
Angular velocity vector	$\Omega = [p \ q \ r]^T$	$\dot{\eta} = [\dot{\phi} \ \dot{\theta} \ \dot{\psi}]^T$
p in terms of $\dot{\eta}$	$p = \dot{\phi} - \dot{\psi} \cdot s\theta$	---
q in terms of $\dot{\eta}$	$q = \dot{\theta} \cdot c\phi + \dot{\psi} \cdot c\theta \cdot s\phi$	---
r in terms of $\dot{\eta}$	$r = \dot{\psi} \cdot c\theta \cdot c\phi - \dot{\theta} \cdot s\phi$	---
Matrix form	$\Omega = w(\eta) \cdot \dot{\eta}$	$\dot{\eta} = w^{-1}(\eta) \cdot \Omega$
5.Angular Acceleration		
Roll acceleration	$\dot{p}$	$\ddot{\phi}$
Pitch acceleration	$\dot{q}$	$\ddot{\theta}$
Yaw acceleration	$\dot{r}$	$\ddot{\psi}$
Vector	$\dot{\Omega} = [\dot{p} \ \dot{q} \ \dot{r}]^T$	$\ddot{\eta} = [\ddot{\phi} \ \ddot{\theta} \ \ddot{\psi}]^T$
6.Angular momentum & Inertia		
Inertia matrix	$I = diag(I_{xx} \ I_{yy} \ I_{zz})$	Varies with orientation
Angular momentum	$H = I \cdot \Omega = [I_{xx} \cdot p \ I_{yy} \cdot q \ I_{zz} \cdot r]^T$	$H = J(\eta) \cdot \dot{\eta}$
Gyroscopic term	$\Omega \times H = \Omega \times (I \cdot \Omega)$	$C(\eta, \dot{\eta}) \cdot \dot{\eta}$
7.Physical constants		
mass	m	0.468 kg
Gravity	g	9.81 m/s ²
Arm length	l	0.225 m
Lift factor	k	2.980*10 ⁻⁶ N.s ² /rad ²
Drag factor	b	1.140*10 ⁻⁷ N.m.s ² /rad ²
Inertia about x and y-axis	I_{xx}, I_{yy}	4.856*10 ⁻³ kg.m ²
Inertia about z-axis	I_{zz}	8.801*10 ⁻³ kg.m ²
Drag Coefficient x/y/z	A_x, A_y, A_z	0.25 kg/s